

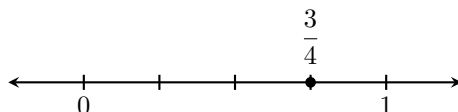
Fractions: Graphing

Graphing Fractions on a Number Line

To graph a **proper fraction**, divide the distance from 0 to 1 into n equal parts, where n is the denominator of the fraction. Then start at 0 and use the numerator to determine the number of places to move to the right.

Example 1. Graph the **proper fraction** $\frac{3}{4}$.

Explanation. Since the denominator is 4, divide the distance from 0 to 1 into 4 equal parts. Then start at 0 and move 3 parts to the right, since the numerator is 3.

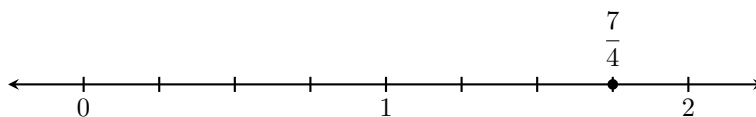


Alt text: A horizontal number line showing the interval from 0 to 1, divided into four equal segments. The integer 0 is labeled at the first tick mark, and the integer 1 is labeled at the fifth tick mark. A solid black dot is placed on the third tick mark from 0 and labeled as three-fourths.

To graph **improper fractions**, we continue the process and divide the distances between each whole number into n equal parts.

Example 2. Graph the **improper fraction** $\frac{7}{4}$.

Explanation. Since the denominator is 4, divide the distance between each whole number into 4 equal parts. Then start at 0 and count 7 parts to the right, since the numerator is 7.



Alt text: A horizontal number line extending from 0 to 2, divided into eight equal segments. The integers 0, 1, and 2 are labeled. A single solid black dot is placed at the seventh tick mark from 0 and labeled as seven-fourths.

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Remark 1. Notice that $\frac{7}{4}$ is equivalent to $1\frac{3}{4}$ since it lies $\frac{3}{4}$ of the way between 1 and 2.

To graph a **mixed number**, first locate the whole number part on the number line. Next, divide the distance between that whole number and the next whole number into n equal parts, where n is the number in the denominator. Then start at the whole number part and graph the fractional part.

Example 3. Graph the **mixed number** $2\frac{1}{5}$.

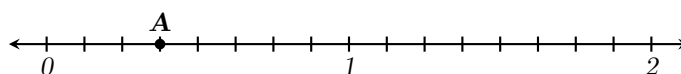
Explanation. Since the denominator is 5, divide the distance between the whole numbers 2 and 3 into 5 equal parts. Then start at 2 and move 1 part to the right, since the numerator is 1.



Alt text: A horizontal number line labeled with the whole numbers 0, 1, 2, and 3. The segment between 2 and 3 is divided into five equal parts. A solid black dot is placed on the first tick mark after 2 and labeled as the mixed number two and one-fifth.

Remark 2. Negative fractions can be graphed in a similar manner.

Problem 1 Identify the position of the letter *A* on the number line.

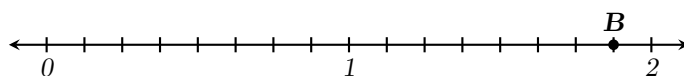


Alt text: A horizontal number line extending from 0 to 2. The distance between each whole number is divided into eight equal parts. Point A is marked with a solid black dot and is located at the third tick mark after 0.

$$A = \frac{\dots\dots\dots}{\dots\dots\dots}$$

Problem 2 Identify the position of the letter *B* on the number line.

Fractions: Graphing



Alt text: A horizontal number line extending from 0 to 2. The whole numbers 0, 1 and 2 are labeled. The distance between each whole number is divided into eight equal parts. Point B is located at the fifteenth tick mark after 0 or equivalently at the seventh tick mark after 1.

Select All Correct Answers:

(a) $1\frac{7}{8}$

(b) $\frac{16}{8}$

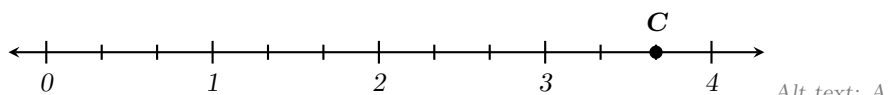
(c) $\frac{15}{8}$

(d) $1\frac{5}{8}$

(e) $\frac{5}{8}$

(f) $\frac{7}{8}$

Problem 3 Identify the position of the letter C on the number line.



Alt text: A horizontal number line extending from 0 to 4. The whole numbers 0, 1, 2, 3 and 4 are labeled. The distance between each whole number has been divided into three equal parts. Point C is labeled with a solid black dot and is located at the second tick mark after 3.

Select All Correct Answers:

(a) $3\frac{1}{2}$

(b) $3\frac{3}{4}$

(c) $\frac{2}{3}$

(d) $\frac{11}{2}$

(e) $3\frac{2}{3}$

(f) $\frac{11}{3}$